STAT/MA 41600

In-Class Problem Set #24: October 14, 2026

1. Consider a random variable X with probability density function $f(x) = \frac{1}{10}e^{-x/10}$ for $x > 0$ and $f(x) = 0$ otherwise.

1a. Find the value of a such that $P(X > a) = 0.75$.

1b. Find the cumulative distribution function of X.

2a. Suppose that Y is a random variable with $f_Y(y) = (k)(y+1)(y-3)^2$ for $-1 < y < 3$, and $f_Y(y) = 0$ otherwise, where k is constant. What is the value of k?

2b. Now drop the square, i.e., define $f_Y(y) = (k)(y+1)(y-3)$ for $-1 < y < 3$, and $f_Y(y) = 0$ otherwise, where k is constant. Can $f_Y(y)$ be a probability density function of a random variable? Why or why not? If so, which value of k should you use? If not, then what fails in this case?

3a. Consider a random variable X with constant probability density function (only) on the interval $(0, 7)$ and otherwise zero. Find $P(|X - 5| < 1/3)$.

3b. Try to generalize this result. Suppose that k is a constant (you may assume that k is bigger than 7). Consider a random variable X with constant probability density function (only) on the interval $(0, k)$ and otherwise zero. Find $P(|X - 5| < 1/3)$.

4. Suppose that the cumulative distribution function of X is $F_X(x) = 1 - e^{-5x/2}$ for $x > 0$, and $F_X(x) = 0$ otherwise.

4a. What is the probability density function of X?

4b. If a is a positive constant, what is $P(X > a)$?

5a. Find the median of the random variable X from Question 1.

5b. Find the median of the random variable X from Question 4.